

# Understanding and Implementing Authenticated Post- Quantum Key Exchange

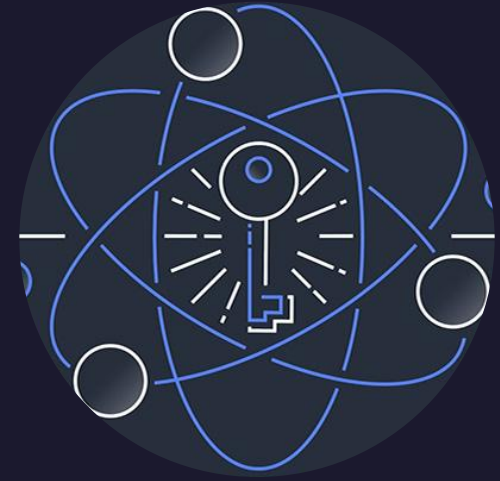
A Practical Study of CRYSTALS-Kyber and EKE



Author: Inês Ribeiro  
Professor: Ricardo Chaves

# Objectives of this Work

- Analyse the challenges that quantum computing poses to classical cryptography
- Study and understand the design of CRYSTALS-Kyber
- Evaluate its main advantages and limitations
- Propose and implement an authenticated EKE-based Kyber construction



# Motivation

Why do we need post-quantum cryptographic algorithms?

## Classical Cryptography

**RSA, ECC**

**Security bad on factoring & discrete log**



## Quantum Threat

**Shor Algorithm**

**Risk to current Systems**



## Need for Post-Quantum Solutions

**Algorithms resistant to Quantum Computers**

**Ex: CRYSTALS - Kyber**



# CRYSTALS - Kyber

## What is Kyber?

**Lattice-based KEM**

**Security based on Module-LWE**

**3 Phases: KeyGen, Encap, Decap**

# Mathematical Structure of Kyber

$$R_q = \mathbb{Z}_q[x]/(x^n + 1)$$

**Key Generation:**

$$\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q}.$$

**Encapsulation:**

$$\mathbf{u} = A^T \cdot \mathbf{r} + \mathbf{e}_1 \pmod{q}$$

$$v = \mathbf{t}^T \mathbf{r} + e_2 + \text{enc}(m) \pmod{q}$$

**Decapsulation:**

$$w = \text{enc}(m) + \mathbf{e}^T \mathbf{r} + e_2 - \mathbf{s}^T \mathbf{e}_1 \pmod{q}.$$

# Kyber - Overview

## Key Generation

**Bob**

Generates a secret **s** and an error **e**

Computes  $t = As + e \pmod{q}$

Sends to Alice:  $pk = (seed_A, t)$

Keeps **s** private

## Encapsulation

**Alice**

Receives Bob's public key: **pk**

Generates random message **m**

Computes ciphertext  $ct = (u, v)$

Derives shared secret **ss** and keeps it private

Sends **ct** to Bob

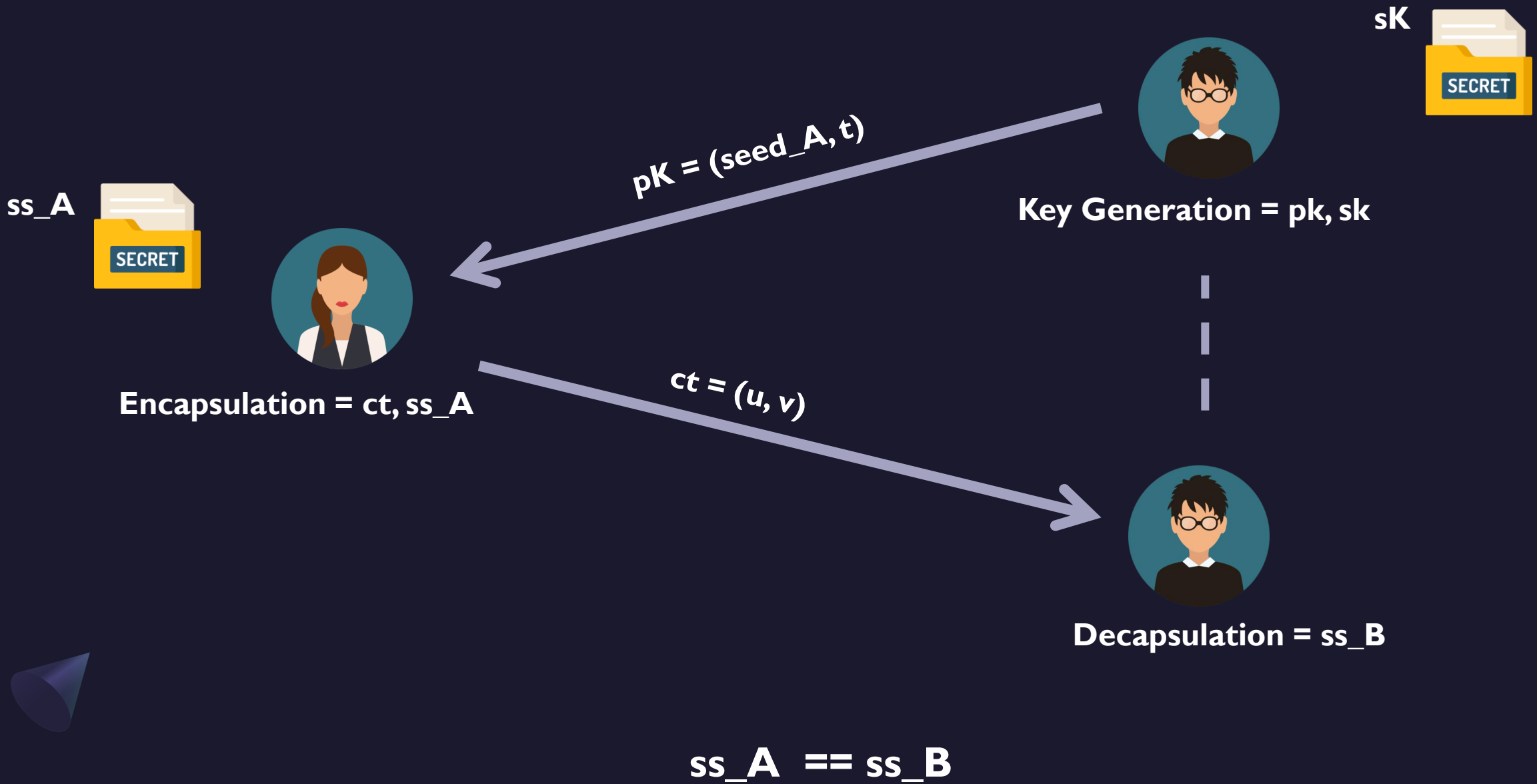
## Decapsulation

**Bob**

Receives ciphertext  $ct = (u, v)$

Uses **s** to cancel **e** algebraically and recovers **m**

Derives same  $ss = KDF(m)$



# The problem of Kyber





# EKE-Kyber

pw = password



$\text{pw\_alice} = \text{PBKDF2}(\text{pw}, \text{salt})$

Alice recovers pk if pw\_alice matches pw\_bob

Encapsulation = ct, ss\_A

$\text{pK} = (\text{seed\_A}, t)$  pw\_bob



$\text{pw\_bob} = \text{PBKDF2}(\text{pw}, \text{salt})$   
Key Generation = pk, sk

pw = password



Decapsulation = ss\_B

$\text{ss\_A} == \text{ss\_B}$



# DEMO



# Conclusion

## What was achieved

**Studied and implemented  
CRYSTALS –Kyber**

**Proposed and implemented  
EKE-Kyber protocol**

**Validated authentication  
through password**

## Limitations

**Security depends on  
password strength**

**No side-channel resistance**

**No formal security proof**

## Future Work

**Formal Security Analysis**

**Side-channel resistant  
implementation**

**Integration in a real  
protocol (ex:TLS)**

# Thank you

